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Technical Note

Estimating the power potential of tidal currents and the impact of power extraction on flow speeds

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ABSTRACT

A simple method for estimating the potential of currents in tidal channels to produce power is presented. The method only requires measurement of the peak tidal volume transport through the channel without turbines, along with a bottom drag coefficient and the channel's dimensions. A recent existing method for estimating potential requires measurements of the undisturbed transport as well as water levels at both ends of the channel to give the head loss. The adaptation of the existing method presented here exploits analytic solutions for the transport and optimal farm drag coefficient and does not require measurement of the head loss. The equations presented allow both the channel's potential and the flow reduction due to power extraction to be estimated using a calculator. Thus the presented method has much of the ease of use of the older KE flux method, but is more reliable as it includes retardation of the flow by the turbines. The presented method can be used for the initial assessment of channels to determine if the additional measurements required to use the existing method are warranted. It can also be used where the headloss in the channel is too small to measure reliably. The presented equations enable the maximum power available to be simply traded off against environmentally acceptable flow speed reduction. The presented method is applied to two example channels. Cook Strait NZ has an estimated potential of 15 GW, while the entrance channel to Kaipara Harbour has a potential between 110 MW and 240 MW.

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1. Introduction

Underwater turbines placed in areas of high tidal currents can contribute to the demand for renewable energy sources. There is already a 1.2 MW tidal stream generator in operation in the high flows of Strangford Lough [1]. To make a significant contribution to electricity demand many turbines will need to be grouped into turbine farms. These farms will likely exploit the high flows through narrow channels. Critical to the development of a tidal turbine farm is having a means to estimate the potential of a tidal channel to produce power. An early method for assessing the potential was to calculate the Kinetic Energy flux through the undisturbed channel (see review [2]). The KE flux method was extremely easy to use, as it only required measurements of tidal currents and a cross-sectional area. However, the KE flux is not related to the power potential of a channel as it does not account for the effect turbines have on the flow along the channel and can significantly under or over estimate a channel's potential. It is also sensitive to which cross-section it is applied [3].

Removing energy from tidal currents affects the flow along a channel by enhancing drag. This retards the flow and limits the power which can be extracted [3]. Garrett and Cummins [4] (hereafter GC05) improved on the KE flux method by developing a simple model which allowed for this retardation of the flow by the turbines. To calculate the potential using their equation required measurements of the undisturbed volume transport along the channel and water levels at both ends of the channel. Here a method is given to calculate a channel's potential without measuring tidal water levels at the ends of the channel. The method exploits an analytic solution to GC05's model given in Vennell [5] (hereafter V10) to calculate the potential from the undisturbed volume transport. Thus the presented method has much of the convenience of the KE flux approach, but is more reliable as it allows for the retardation of the flow due to power extraction. Flow retardation may have environmental impacts along much of the channel, such as reduced tidal exchange or sediment transport. Whether potential is estimated using GC05's equation or those presented here V10's analytic solution can also be used to estimate the flow speed reduction due to the farm. This enables trading off the benefits of power production against environmentally acceptable flow speed reduction. This work aims to present methods for

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estimating potential and flow speed reduction which are useful in the initial assessment of potential farm sites.

Both the GC05 method and the presented method require measurement of the volume flux along the undisturbed channel. Volume flux is easily measured using vessel mounted ADCPs by multiple vessel crossings of the channel throughout a tidal cycle (e.g. [6,7]). The GC05 method requires an additional measurement, the head loss amplitude (see Eq. (2)). Where tide gauge measurements from both ends of the channel are available to estimate the head loss, using the existing GC05 method is straight forward. The presented method is useful where tide gauge measurements at both ends of the channel are not available. It is particularly useful for small channels where the head loss may only be a few centimeters and too small to measure reliably from the difference of tide gauge measurements, which may be of limited accuracy. Thus the method is useful for the initial assessment of small channels which might power an isolated community. For a channel connecting the open ocean to an inlet or lagoon there is a modified form of GC05 [8] which only requires measurements of the tidal range in the ocean and the lagoon area. The potential and flows obtained from the modified GC05 model will be compared to those from the presented method for the less common situation where only flow measurements are available for a channel connecting to a lagoon.

2. Background

The KE flux averaged over a tidal cycle is easily calculated from

$$\mathrm{KE}_{\mathrm{flux}} = \frac{\rho}{2A^2}\frac{4}{3\pi}U_{0\mathrm{UD}}^3 \tag{1}$$

where ρ is seawater density, A is the cross-sectional area of the channel and $U_{0\mathrm{UD}}$ is amplitude of the tidal transport along the channel without turbines, i.e. the peak volume transport in the undisturbed channel. KE flux is easy to calculate as it only requires estimates of $U_{0\mathrm{UD}}$ and A at the farm's location to calculate the "potential". This transport could be obtained from current measurements or from the output of a hydrodynamic model of the region. For the two example channels discussed in detail in Section 6 the KE flux underestimates the potential by up to an order of magnitude. If the KE flux consistently underestimated potential it would be useful as a conservative estimate, however for shorter channels (such as a channel similar to the first example but half as long) KE flux can significantly overestimate the potential and thus is not a reliable method. The KE flux is not related to the power potential of a channel because it does not allow for the effect that the turbines have on the flow along the channel.

A result from GC05's more realistic model for a channel was an equation for the maximum power which can be lost to a tidal turbine farm, i.e. the channel's potential $\overline{P}_{\max}$. It assumes that turbines occupy the whole cross-section and that there are no hydrodynamical or electro-mechanical losses of energy. Thus the potential is an upper bound for the power which can be derived from the channel. GC05's surprisingly simple equation for a channel's potential is

$$\overline{P}_{\max} = 0.22\rho g\zeta_0 U_{0\mathrm{UD}} \tag{2}$$

where g is the acceleration due to gravity and ζ_0 the amplitude of the water level difference or head loss between the ends of the channel. Across the range of possible channel dynamical balances the coefficient of the RHS in Eq. (2) varies between 0.21 and 0.24. Thus within 10% the equation can be applied to the full range of dynamical balances from shallow channels dominated by bottom friction, to large tidal straits where flow inertia dominates.

Though GC05 gives a more realistic estimate of a channel's potential Eq. (2) than the KE flux, it is more complicated to use as it also requires measurement of the amplitude of the tidally varying head loss between the ends of the channel, ζ_0. This could be obtained from tidal gauges installed at both ends of the channel, or from the output of a regional hydrodynamic model. However there may not be tidal data from both ends of the channel nor model output available. Also the head loss ζ_0 may be too small to measure reliably. In these cases Eq. (2) cannot be used.

GC05's result was derived from a numerical solution to a one-dimensional model for a channel with turbines forced by an oscillating head loss between the ends with amplitude ζ_0. In principle it would be possible to use GC05's numerical model iteratively without turbines to adjust ζ_0 until the model gives the measured undisturbed transport $U_{0\mathrm{UD}}$. Once the required ζ_0 is found it can then be used in Eq. (2) to give the potential. This process would require effort to set up and use their numerical model. The adapted GC05 approach presented here in Section 3 uses an approximate analytic solution to GC05's model and an analytic expression for the farm's optimal drag coefficient, given in V10, to replace GC05's numerical model in this process. The analytic solution is used twice, firstly for the undisturbed channel to give the head loss amplitude ζ_0, then secondly for the disturbed channel to give its potential. This reduces expressions for ζ_0, the potential and the associated flow reduction to equations which could be evaluated with a calculator.

Estimating the potential of a channel connecting the open ocean to a lagoon or tidal estuary requires a subtly different form of Eq. (2). For tidal inlets or lagoons like Kaipara Harbour in Fig. 1a, Blanchfiled et al. [8] showed the potential can be estimated within approximately 15% by

$$\overline{P}_{\max} = 0.21\rho g\eta_{01}U_{0\mathrm{UD}} \tag{3}$$

where η_{01} is the amplitude of the tidal elevation in the ocean. The difference arises because Eq. (2) assumes the tidal amplitudes in

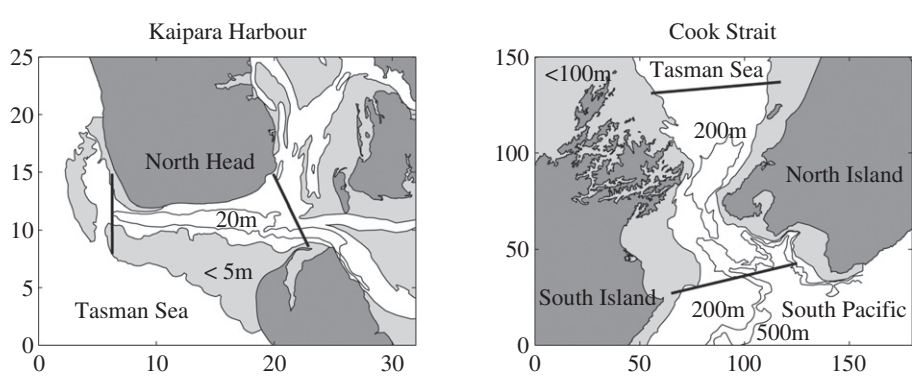

Fig. 1. Maps of two example channels. Axes are in km and thick dark lines demark the ends of the channel used to estimate their tidal current power potential in Table 1.

the water bodies connected by the channel are unaffected by power extraction. However, from mass continuity the tidal elevation amplitude within the lagoon is dependent on the volume flux into the lagoon. Thus power extraction, which reduces flows along the channel, also reduces tidal amplitudes within the lagoon. For smaller channels like Kaipara Harbour the difference in water level between the ends of the channel is typically smaller than the tidal amplitude, i.e. $\zeta_0 < \eta_{01}$. As a result GC05's channel potential (Eq. (2)) will typically underestimate the potential of a channel connected to lagoon (Eq. (3)). Thus both GC05 and the presented method will generally yield conservative estimates of the potential of these channels.

3. Estimating the potential of a channel

3.1. Model and useful results

In GC05's one-dimensional model for a narrow channel, tidal flows are driven along the channel by differences in water levels in the large water bodies at the ends of the channel. Their model assumes velocity is uniform across the channel's cross-section, the channel is short [9,10]. Their model allows for a variable cross-section. For simplicity a uniform rectangular cross-section will be used here. The momentum balance for their channel with turbines can be expressed as

$$\frac{\partial U}{\partial t} = -\frac{gA}{L}[\eta_2 - \eta_1] - \frac{A}{L}\left[\frac{L}{h}C_D + C_F\right]\frac{|U|U}{A^2} \tag{4}$$

where U is the volume transport along the channel, A its cross-sectional area, L its length and h its depth. η_1 and η_2 are the tidal water levels at the ends of the channel. These water levels are assumed to be unaffected by the turbines within the channel. C_D is the drag coefficient for background bottom friction based on the channel's horizontal area and C_F is the drag coefficient due to power extraction by the farm based on the channel's cross-sectional area A.

The transport along the channel at any time can be written in the form $U = U_0\sin(\omega t + \phi_u)$ where U_0 is the amplitude of the transport along the channel with turbines installed in $m^3\,s^{-1}$, ω is the tide's angular frequency and ϕ_u is the phase of the transport. With this the average power lost by flow to the turbines is

$$\overline{P}_{lost} = \frac{4}{3\pi}\frac{\rho C_F}{A^2}U_0^3 \tag{5}$$

where the $4/3\pi$ factor results from averaging over a tidal cycle.

V10 gives an approximate analytic solution to Eq. (4) which differs from the numerical solution by less than 5%. Such a solution is more than adequate given the model assumptions. V10's solution for the amplitude of the tidal transport is

$$U_0 = U_I\sqrt{\frac{\sqrt{4\lambda^2+1}-1}{2\lambda^2}} \tag{6}$$

The non-dimensional constant λ is defined by

$$\lambda = \frac{8\alpha}{3\pi}\left[\frac{L}{h}C_D + C_F\right] \tag{7}$$

and the dimensionless constant

$$\alpha = \frac{U_I}{\omega L A} \tag{8}$$

is also useful as it indicates the dynamical balance within the channel. In a channel with small α bottom friction is unimportant, while large α indicates bottom friction dominates the flow's inertia. In Eq. (6) the constant U_I is the transport's scale. U_I has a physical interpretation as the amplitude of the transport that would occur in the channel if there were no bottom friction or turbines, i.e. the transport amplitude as $\lambda \to 0$.

As a farm is expanded its drag coefficient will increase. The combined effect on the power lost (Eq. (5)) of increasing C_F and the associated decrease in transport (Eq. (6)) is to create a power curve for which there is a peak that defines the maximum power that can be lost by the flow to the turbines, GC05. The power lost at this peak in the power curve, $\overline{P}_{max}$, is the channel's potential. $\overline{P}_{max}$ occurs at an optimal farm drag coefficient C_F^{peak}. Power curves for the two examples are given in Fig. 2. These curves have been scaled by each channel's potential and optimal drag coefficient in Table 1. These scaled curves are almost identical, demonstrating the near similarity across the dynamical range discussed in V10.

The optimal value of C_F can be found by substituting Eq. (6) into Eq. (5) differentiating, then setting the result equal to zero and solving for C_F. The resulting equation is too complicated to be useful. However, V10 showed that, to a good approximation, the farm's optimal drag coefficient can be expressed as

$$C_F^{peak} \approx 2\frac{L}{h}C_D + \frac{3\pi\sqrt{2}}{8\alpha} \tag{9}$$

The potential is found by evaluating Eq. (5) and Eq. (6) at Eq. (9), i.e.

$$\overline{P}_{max} = \frac{4}{3\pi}\frac{\rho C_F^{peak}}{A^2}\left(U_0^{peak}\right)^3 \tag{10}$$

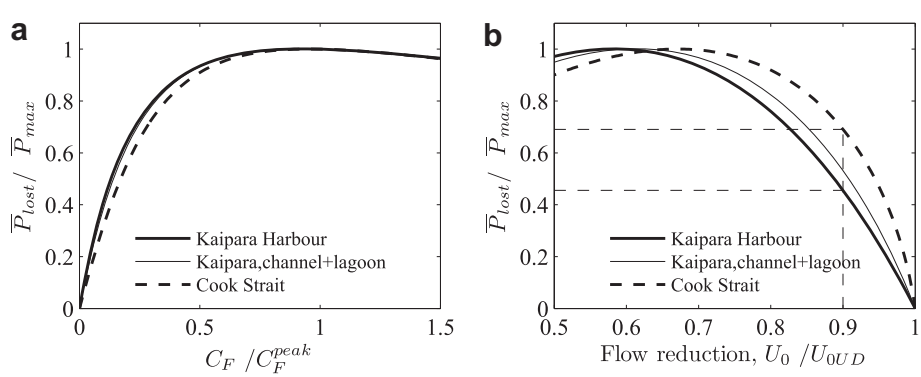

Fig. 2. (a) Power curves for the two example channels normalized by optimal farm drag coefficient C_F^{peak} and potential $\overline{P}_{max}$. The power curve for the Kaipara channel + lagoon model is indistinguishable from the Kaipara channel model. (b) Effect of power extraction on flow speed reduction for the two examples. Thin dashed lines indicate the power lost to the turbines at 90% of the undisturbed flow speeds.

Table 1
Calculation of tidal power potential for two example channels. For both channels semi-diurnal tides dominate, $\omega = 1.4 \times 10^{-4}$ radians/s, $\rho = 1025$ kg m^{-3} and $C_D = 0.0025$. The transport units are in Sverdrups, SV $= 10^6$ m^3 s^{-1}. The results for a channel connected to a lagoon for the Kaipara Harbour are based on a lagoon area $A_L = 580$ km^2 and oceanic tidal amplitude $\eta_{01} = 1.05$ m giving $\beta_L = 3.6$. Power available to a given number of optimally tuned turbines is based on 10 rows of turbines occupying a total of 10% of the channel's cross-section. Turbine numbers are based on a turbine with a pair of 16 m diameter blades.

	Kaipara Harbour		Cook Strait
Length, L	15 km		100 km
Depth, h	25 m		150 m
Width, W	2.5 km		25 km
Cross-sectional area, A	63,000 m^2		3.8 km^2
	Channel	Channel + Lagoon	
Observed undisturbed transport amplitude, U_{0UD}	0.11 SV	0.11 SV	4.1 SV
Undisturbed velocity amplitude $= U_{0UD}/A$	1.7 ms^{-1}	1.7 ms^{-1}	1.1 ms^{-1}
$D = 8C_D/3\pi\omega A\ h$	9.7×10^{-6} m^{-3}s	–	2.7×10^{-8} m^{-3}s
Transport scale, U_I, Eq. (13)	0.15 SV	–	4.1 SV
Head loss amplitude, ζ_0, Eq. (14)	0.5 m	–	1.6 m
$\alpha = U_I/\omega LA$ or $\alpha^* = g\eta_{01}/\omega^2 L^2$	1.2	2.3	0.079
Optimal farm drag coefficient, C_F^{peak} (9) or (18)	4.4	7.8	24
λ_{peak} from Eq. (7)	5.8	7.7	1.8
Transport amplitude at peak, U_0^{peak} (6)	0.06 SV	0.07 SV	2.7 SV
Velocity amplitude at peak $= U_0^{peak}/A$	0.97 ms^{-1}	1.05 ms^{-1}	0.72 ms^{-1}
Flow reduction at peak, U_0^{peak}/U_{0UD}	0.57	0.60	0.66
Tidal current potential, $\overline{P}_{max}$ (10)	110 MW	240 MW	15 GW
Potential from GC05 approx. formula, (2) or (3)	120 MW	240 MW	14 GW
Average KE flux in undisturbed channel (1)	66 MW	66 MW	1.1 GW
Power available to 10 rows occupying 10% of the cross-section from V10	37 MW	46 MW	0.8 GW
Number of optimally tuned turbines required	150	150	9000

where U_0^{peak} is the along channel transport at the peak in the power curve in Fig. 2b, i.e. the retarded transport when fully realizing a channel's potential.

3.2. Calculating ζ_0 and U_I from tidal measurements at both ends

The potential cannot be estimated from Eq. (10) without knowing the transport scale U_I in Eq. (6). If there are tide gauge measurements at both ends of the channel or detailed hydrodynamical model output is available then U_I can be estimated by the following. Tidal analysis of the measured or modelled water levels at the ends of the channel can be used to calculate the tidal amplitudes at the ends, η_{01} and η_{02}, and their corresponding phases, ϕ_1 and ϕ_2. The amplitude and phase of the sinusoidal headloss between the ends of the channel in Eq. (4) can be written as $\eta_1 - \eta_2 = \zeta_0 \sin(\omega t + \phi_g)$ where

$$\zeta_0 = \sqrt{\eta_{01}^2 - 2\eta_{02}\eta_{01}\cos(\phi_2 - \phi_1) + \eta_{02}^2} \tag{11}$$

and $\phi_g = \tan^{-1}(\eta_{01}\sin\phi_1 - \eta_{02}\sin\phi_2)/(\eta_{01}\cos\phi_1 - \eta_{02}\cos\phi_2)$. From this the transport scale U_I is easily calculated using

$$U_I = \frac{g\,A\,\zeta_0}{\omega\,L} \tag{12}$$

However, this paper addresses the case when the amplitudes and phases of the tide at the ends of the channel are not readily available or their differences are too small to be reliably measured.

3.3. Calculating U_I and ζ_0 from the undisturbed volume transport U_{0UD}

The transport solution (6) can be applied to the undisturbed channel and used to estimate the transport scale U_I from the measured transport along the undisturbed channel. Rearranging (6) for U_I with $C_F = 0$ and simplifying gives

$$U_I = U_{0UD}\sqrt{D^2 U_{0UD}^2 + 1} \tag{13}$$

where $D = 8C_D/3\pi\omega Ah$. For large tidal straits bottom friction is unimportant, i.e. $D \to 0$. Thus for these channels $U_I \approx U_{0UD}$ as expected. Once U_I has been calculated from Eq. (13) ζ_0 is easily calculated from Eq. (12)

$$\zeta_0 = \frac{\omega U_I\,L}{gA} \tag{14}$$

4. Calculating the potential from the undisturbed transport U_{0UD}

Given the above equations the steps to find the potential from the undisturbed transport are as follows.

1. Using the measured undisturbed transport U_{0UD}, a bottom friction coefficient C_D, the dimensions of the channel and the tide's angular frequency ω firstly find D then U_I using (13).
2. With U_I calculate α from Eq. (8).
3. Using α and C_D calculate the optimal farm drag coefficient C_F^{peak} using Eq. (9).
4. Use C_D, C_F^{peak} and α to give the optimal λ_{peak} Eq. (7)
5. Use λ_{peak} with U_I to calculate the amplitude of the retarded along channel transport at optimal drag coefficient U_0^{peak} using Eq. (6).
6. Finally use $C_F^{peak}$and U_0^{peak} in Eq. (10) to give the potential $\overline{P}_{max}$.

A shorter way to calculate the potential is to use U_I from step 1 to give ζ_0 using Eq. (14). ζ_0 along with U_{0UD} could then be used in GC05's approximate formula Eq. (2) to give the potential. The advantage of the longer six-step method is that it also gives an estimate of the along channel transport at the peak in the power curve. The ratio U_0^{peak}/U_{0UD} is the reduction in flow speed along the channel required to fully realize a channel's potential, which is useful in assessing the environmental impacts of power extraction. Fig. 3 outlines the steps and the flow of information for the six-step method and the shorter method using GC05's equation.

5. Results for a channel connected to a lagoon

GC05's model, on which the presented method is based, assumes the elevations at the ends of the channel are unaffected by power extraction. However, for a channel connecting an inlet or lagoon to the ocean, the tidal elevation within the lagoon, η_2, is related to the transport along the channel by mass continuity, i.e.

$$A_L \frac{\partial \eta_2}{\partial t} = U \tag{15}$$

where A_L is the surface area of the lagoon. Incorporating this into the momentum Eq. (4) also leads to an approximate analytic solution given by [11]. Their solution recast in terms of transport amplitude is

$$U_0 = \frac{gA\eta_{01}}{\omega L}\sqrt{\frac{\sqrt{4\lambda^2 + (1-\beta_L)^4} - (1-\beta_L)^2}{2\lambda^2}} \tag{16}$$

where $\beta_L = gA/L\omega^2 A_L$ is a lagoon parameter and η_{01} is the amplitude of the oceanic tide. The non-dimensional constant λ is still defined by Eq. (7) but with α is replaced by α^*, which is defined by

$$\alpha^* = \frac{g\eta_{01}}{\omega^2 L^2} \tag{17}$$

For a channel with a lagoon the optimal drag coefficient for a farm given in V10 is modified to

$$C_F^{peak} \approx 2\frac{L}{h}C_D + \frac{3\pi\sqrt{2}}{8\alpha^*}(1-\beta_L)^2 \tag{18}$$

This along with Eq. (16) can be used to estimate the potential of the channel connected to a lagoon using Eq. (10). This will be compared to that from the presented method applied to the Kaipara Harbour.

6. Example channels

6.1. Kaipara Harbour

Kaipara Harbour is located on the north west coast of New Zealand's North Island, Fig. 1a. This large estuary has an area at high tide of 947 km^2 with 409 km^2 exposed at low tide. The Harbour is dominated by semi-diurnal tides with tidal ranges of between 1.52 and 2.68 m. The Harbour's tidal prism, the volume of water entering the Harbour during a flood tide, is 1.99 km^3 at spring tides and 1.13 km^3 at neap tides [12]. The 15 km long and 2.5 km wide entrance channel averages around 25 m deep with a maximum depth of 50 m, Table 1. The entrance channel terminates in a shallow ebb tidal bar at its seaward end. The channel's value of α is near 1, indicating that the flow's dynamical balance is near the limit where bottom friction dominates and inertia is unimportant. Work by [13] indicates typical peak tidal channel velocities are around 1.8 m/s. Based on an average of spring and neap tidal prisms [12] cross-sectional average velocities are estimated at 1.7 ms^{-1}. This figure is

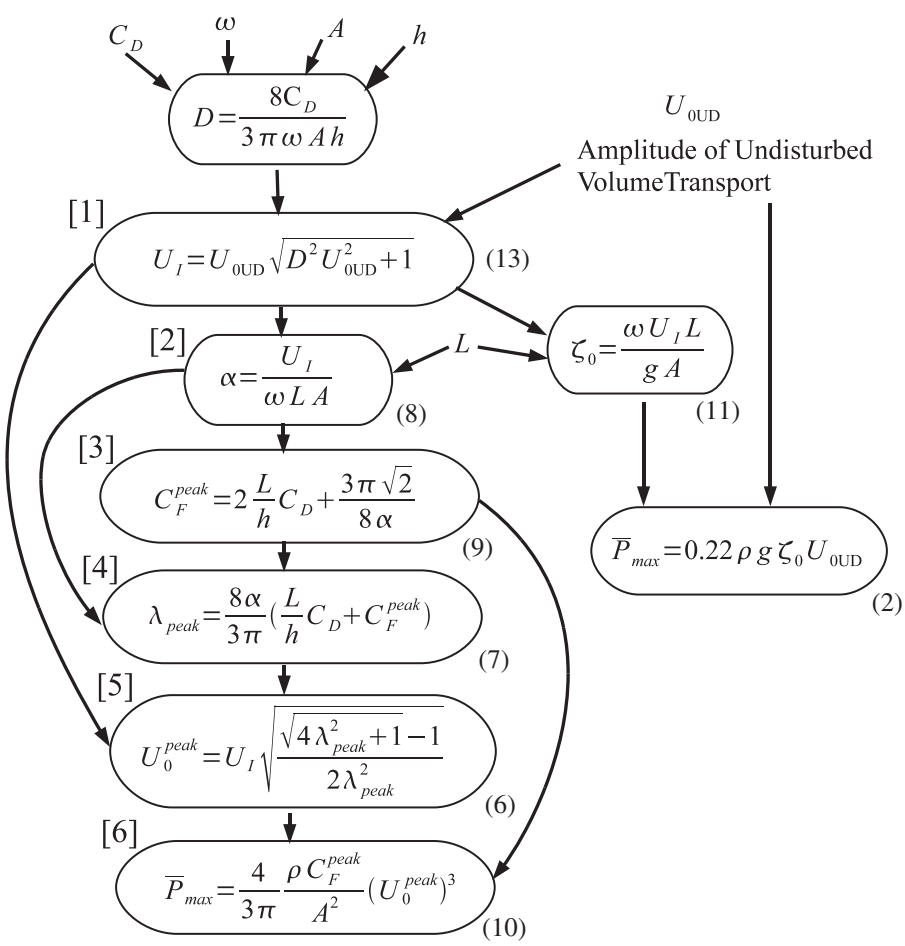

Fig. 3. Calculation flow or influence diagram for the six-step estimation of potential of a channel from the undisturbed transport and using GC05's Eq. (2). Square bracketed numbers give the step and round brackets are the equation numbers.

used to give the undisturbed volume transport used in the presented method. The channel with lagoon solution for the undisturbed channel Eq. (16) gives a very similar velocity based on an oceanic tidal amplitude of 1.05 m and a mid-tide estuary area of $A_L = 580\ km^2$ [12] in Table 1.

The Kaipara Harbour entrance channel's estimated potential for an average tidal range is 110 MW with GC05's approximate Eq. (2) giving a similar value of 120 MW. The channel with lagoon model, Eq. (16) with Eq. (18), gives a higher potential of 240 MW. However, the channel and lagoon model ignores any energy losses due to bottom friction in the large shallow harbour where around 40% is exposed at low tide. Thus the Kaipara Harbour Channel's potential will likely lay in the range 110 MW–240 MW. Both models show that an impact of installing sufficient turbines to realize the channel's potential would be to reduce flows to around 60% of their undisturbed velocities.

6.2. Cook Strait

Cook Strait separates New Zealand's two main islands. The dominant M2 tide rotates as a Kelvin wave anti-clockwise around the NZ land mass resulting in tides which are almost in anti-phase between the northern and southern ends of the Strait [14]. This large phase difference drives strong flows through the Strait [15]. The vertical cross-section of the measured amplitude of semi-diurnal tidal currents by [6] were used to estimate the volume transport at right angles to the east-west ADCP measurement transect. Across the Strait's narrowest cross-section this gives average velocities of 1.1 ms^{-1}, though there are weaker flows on the western side and velocities in excess of 2 ms^{-1} on the eastern side of the Strait. The tidal amplitudes and phase of tidal elevation at the ends of the designated Strait estimated from the coarsely contoured coastal tide gauge data in [15] are around 0.65 m/140° and 1.1 m/270°. Using Eq. (11) these give a ζ_0 similar to that estimated from the transport in Table 1. The tidal current potential of the Strait is huge, 15 GW, a very similar value is given by GC05's approximate Eq. (2). This estimate is based on the narrowest width of the Strait. The sensitivity to channel width explored in Section 7 indicates that increasing the width from 25 km to 35 km would reduce the Strait's potential to 11 GW.

The estimates of potential in Table 1 are based on an average tidal current amplitude, that of the dominant M2 semi-diurnal tide. The potential will vary throughout a month due to the effects of the spring neap cycle which results from the combined effects of lunar and solar tides. As power varies with transport cubed, the values in the table based on average current will underestimate the power potential averaged over a full spring neap cycle. For example if the solar semi-diurnal tide is 40% the size of the semi-diurnal lunar tide, then the average power over the resultant spring neap cycle will be 36% higher than the potentials given in Table 1 for the lunar tide alone.

7. Sensitivity of estimated potential to input values

The values U_{0UD}, C_D and the channel's length, width and depth used to estimate the potential may be only approximately known. Thus it is important to vary the input values over a reasonable range to quantify the sensitivity of the calculated potential to these inputs. This is done for the two examples in Fig. 4. The potential is sensitive to U_{0UD} with a 50% larger U_{0UD} giving 2.8 times the potential in Kaipara Harbour. Cook Strait is slightly less sensitive to any underestimate of U_{0UD}. This sensitivity is expected given the dependence of potential on the transport cubed in Eq. (10). Fig. 4b shows the sensitivity of the potential to changes in the bottom friction drag coefficient, C_D around the typical value of 0.0025 used in Table 1. For the examples potential is relatively insensitive to C_D. In Cook Strait the choice of C_D has almost no affect on the potential as bottom friction is unimportant to its dynamical balance.

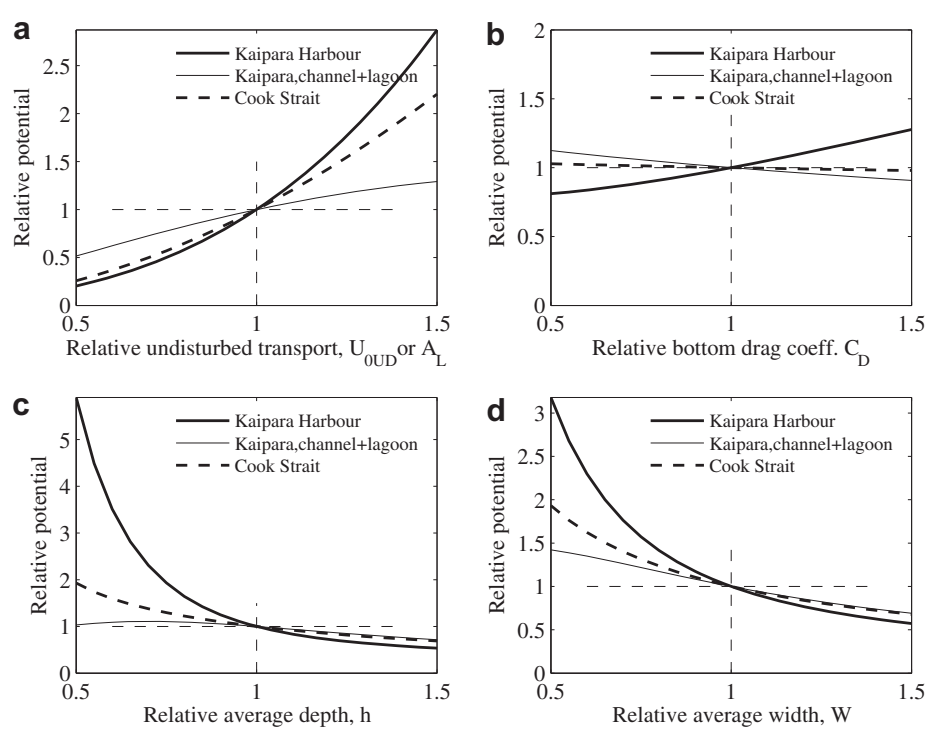

Fig. 4. Sensitivity of potential to variation in the data used to estimate it. The horizontal axes are the fraction of the values given in Table 1 for each example. (a) Effect of changing U_{0UD}, or, for channel + lagoon model, the effect of changing lagoon surface area, A_L. (b) Effect of changing C_D. (c) Effect of changing average water depth, h. (d) Effect of changing average width, W.

Though exaggerated, any uncertainty in the channel's depth mirrors any uncertainty in C_D in Fig. 4c, where a 50% smaller average depth for Kaipara Harbour channel increases the potential by 5 times for fixed U_{0UD}. There is much less variation in potential due to uncertainty in the average depth of Cook Strait, which is encouraging given the wide variation in water depth over the designated channel in Fig. 1b. The potentials have a similar sensitivity to average channel width in Fig. 4c as they do to depth.

Increasing the length, while keeping U_{0UD}, C_D and the other dimensions constant, proportionately increases the channel's potential, so that a 50% longer channel has 50% more potential. This is easily inferred from GC05's (2). Flow in the undisturbed channel is driven by the slope in the water's surface due to the differences in water elevation at the ends. A given U_{0UD} requires a given slope, thus increasing the length for a fixed U_{0UD} proportionately increases ζ_0 and hence the potential given by (2).

The thin solid curves in Fig. 4 give the sensitivity to changes in estimates of measurements required for the channel + lagoon model of Kaipara Harbour. The channel + lagoon model does not require measurement for U_{0UD}, however, Fig. 4a shows it is less sensitive to uncertainty in lagoon area than the presented method is to measurement of U_{0UD}. The channel + lagoon has a similar sensitivity to the bottom friction drag coefficient, while being less sensitive to uncertainties in channel depth and width.

8. Assessing the impact of the farm on the flow

Whether ζ_0 is available from tide gauges, model output or estimated from flows as outlined in Section 4, the equations of Section 3 can be used to trade off power against environmentally acceptable reduction in flow speeds.

For channels where tide gauge data or model output is available the first step is to calculate U_I from ζ_0 (Eq. (12)). For the cases of tide gauge data or using only flow data given in Section 4 the subsequent steps are the same. With U_I and for a range of farm drag coefficients C_F from 0 to C_F^{peak} calculate the transport using (6). This in turn can be used to give the power lost for each C_F using Eq. (5), yielding a table of C_F, U_0 and $\overline{P}_{lost}$. These tabulated values for the two examples are plotted in Fig. 2b. The channel + lagoon model for Kaipara gives a similar trade off curve to that from the presented method. Fig. 2b demonstrates that if the flow in the Kaipara Harbour entrance channel is to be maintained at 90% of its undisturbed value then the maximum power available from the channel is 48% of its potential. For Cook Strait a maximum of 68% of its potential may be realized from the same flow speed reduction.

9. Discussion

The presented method summarized in Fig. 3 provides a means to estimate the potential of a channel where only very basic information about currents is available, thus the method is well suited to initial assessments. It can be used to determine whether a channel warrants further measurements or the hydrodynamical model development required to estimate the potential using GC05's (Eq. (2)). Though moderately sensitive to the values of transport and the channel's average width and depth, the method can also be used to quantify the range of the potential due to uncertainties in the values used to calculate it. The method is applied to a channel's representative width and depth, thus is less sensitive to cross-sectional area than the KE flux. The method's sensitivity to cross-sectional area is inherent in the power calculation (Eq. (10)) and the GC05 model (Eq. (4)). GC05's equation for the approximate potential (Eq. (2)) has the advantage of being only linearly dependent on uncertainties in the measured values of U_{0UD} or ζ_0. However for some channels the head loss amplitude ζ_0 may not be available or be too small to be reliably determined from differences between tide gauge measurements at the ends of the channel.

For channels connecting the ocean to a lagoon the adaptation of the GC05 model by [8,11] can be used to estimate its potential. This adaptation does not require measurement of U_{0UD}, but requires oceanic tidal amplitude and lagoon area. For channels where these measurements are not available, but U_{0UD} is, then the presented method could be used. Flow reduction due to power extraction in the channel reduces the volume of water flowing into a lagoon. This reduction reduces the tidal amplitude within the lagoon. For example when realizing Kaipara's potential the tidal amplitude in the lagoon is only 0.8 m, compared to 1.05 m in the ocean. The resulting larger headloss between the ends of the channel leads to a higher potential estimated from the channel + lagoon model than the presented method, which does not allow for the effect of power extraction on tidal range at the lagoon end of the channel. Thus the presented method can be used as a conservative estimate of the potential of a channel connected to a lagoon when $\zeta_0 < \eta_{01}$. In addition the channel + lagoon model does not allow for bottom frictional energy loss within the lagoon and may over estimate a channel's potential. Thus the channel's potential will likely lie between that estimated by the presented method and that from the channel + lagoon model. The channel + lagoon model does have the advantage of having similar or lower sensitivity to uncertainties in the required measurements.

While having a simple method to estimate a channel's potential is useful, it is also important to be able to estimate the associated flow speed reduction for use in assessing the environmental impacts of power extraction. Whether the potential is calculated from transport alone using the presented method or using the transport and two water levels using GC05's (Eq. (2)) the Eqs. (5)–(7) can be used to easily assess the maximum power which can be extracted from the channel for a given environmentally acceptable flow reduction. For a channel connected to a lagoon the flow reduction curve estimated from the presented method is similar to that from the channel + lagoon model (Fig. 2b).

A channel's assessed potential is the maximum power which can be lost by the flow to the turbine farm. The farm's electrical power output will be significantly lower than this upper bound due to a number of factors. The estimated potential assumes that tidal turbines occupy the entire cross-section of the channel. Gaps between turbines will almost always be needed for navigation of vessels and marine life. The understanding of the power available from a farm where some flow bypasses the turbines through gaps within rows of turbines is limited [5,16,17]. V10 showed that there are two extreme ways to realize most of a tidal channel's potential, both requiring the turbines to be tuned for a particular channel and cross-sectional turbine density. Either the turbines must occupy most of the channel's cross-section, or, if this is not possible, then a very large number of turbines in many rows spread along the channel are required. Table 1 gives estimates based on V10 showing how much of the potential is available to a farm with ten rows of optimally tuned turbines whose blades are permitted to fill only 10% of the cross-section. For Kaipara this makes 37 MW available, around 1/3 of the its potential, and requires 150 turbines. Due to weaker flows in Cook Strait this farm would make only 0.8 GW available, i.e. just 5% of its potential, and requires 9000 turbines. Increasing the fraction of the cross-section filled by the turbines makes the same power available from fewer turbines, an aspect explored more thoroughly in [18]. This leads to the conclusion that the power available from an affordable number of turbine rows with gaps is likely to be much less than a channel's potential. Compounding this is that the flow loses some of its energy to drag on the turbine support structures. Thus electrical output will be further reduced by these losses and the less than perfect electro-mechanical efficiency of the turbines, as well as any transmission or conversion losses.

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